Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 6.1 · A collection of derivatives

Functions are considered

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. Specify the area of definition of each of these functions.

2. Calculate their derivatives by naming each time the rule used. g, first keep the two terms of the product rule; for h, write the numerator before reducing it.

3. Control the result concerning g developing its expression before drifting.

Exercise 6.2 · A complete picture

The function f is defined on [0; 4] by $f(x) = x^3 - 6x^2 + 9x$ + 1. We look for its minimum values and maximum in this interval.

1. Calculate and factorize $f\prime(x)$, then study his sign on [0; 4].

2. Draw up the table of variations of the f indicating the values at the bounds and at the points where the Derivative from the void.

3. Give the minimum and maximum of f on the interval, as well as all the apses where they are Affected.

Exercise 6.3 · A quotient

For any real strictly positive x, we're laying $f(x) = x + \frac{4}{x}$.

1. Studying variations in f on]0; + ∞[. Present the factors that determine the sign of derivative.

2. Deduct the smallest value of $x + \frac{4}{x}$, then write an inequality valid for all $x > 0$. Specify exactly the case of equality.

Exercise 6.4 · Compare two curves

Functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. We note $\mathcal{C}_f$ and $\mathcal{C}_g$ their curves in a same coordinate system.

1. Determine the intersection points of these two curves.

2. Study their relative position on $\mathbb{R}$ by choosing a suitable method. Translate your result by intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

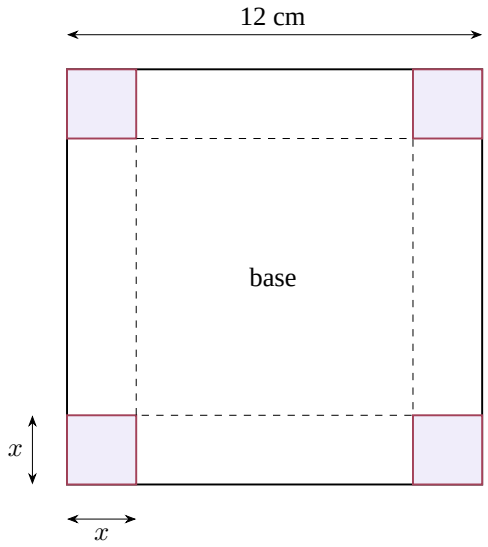
Exercise 6.5 · One package

A box without lid is made from a square sheet of 12 cm side.
in each of the four corners a square side x cm, then the remaining edges are raised.
The leaf is neglected.

1. Draw the pattern and express the height and the two dimensions of the base of the box. Specify the values of x which give a non-zero volume box.

2. Express Volume $V(x)$ in cubic centimetres. Determine the dimensions that make this volume maximum.

3. Check the values at the two bounds of the domain and give the exact maximum volume.



The four coloured areas are removed; the dots indicate the folds. The drawing represents a illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without extremum

A student says: "If the derivative cancels in one point, the function necessarily has an extreme "We are studying the function. $f(x) = x^3$ to test this statement.

1. Calculate $f\prime(0)$.

2. Compare to $f(0)$ values of $f(x)$ for real x negative then positive, as close to 0 as We want to.

3. Conclude on the existence of a local extremum in 0 and on the proposed statement.

Exercise 6.7 · Compare Strategies

We want to show that, for all real x, $x^2 - 6x + 12 \geq 3$.

1. Propose a demonstration by showing a square.

2. Propose a second demonstration by studying variations in a function.

3. Specify the case of equality. Compare the two methods: which one asks here the least calculation?

Fiscal year 6.8 · Deferred review

Reciprocal function $f(x) = \frac{1}{x}$ a for field $D = \mathbb{R} \setminus \{0\}$. A student calculates $f\prime(x) = -\frac{1}{x^2} < 0$ and concludes that f is decreasing on all D.

1. Identify the missing hypothesis in the application of the theorem linking derivative sign and changes.

2. Indicate the two intervals on which its conclusion is correct.

3. Using Numbers -1 and 1, build a counter-example to its conclusion on all D.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 12 p. 156, Questions 1 and 2. Calculating the difference, studying its sign and translate it into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. Give an interval for each function where it is both negative and increasing; justify both properties separately.